

USPTO Non-Provisional Utility Patent Filing Checklist

HEALTH-BAND Neuro — Application Package

Inventor: Srikanth Patchava

Assignee: Embedded Operating Systems Research Foundation (EoS)

Priority Application: U.S. Provisional No. ⁶⁴/_{073,334} (Filed May 23, 2026)

Filing Deadline: May 23, 2027 (12-month statutory deadline)

Target Filing Window: November 2026 – March 2027

Required Documents for USPTO Non-Provisional Filing

1. Application Data Sheet (ADS) — USPTO Form PTO/AIA/14

- ☐ Inventor name(s) and residence
- ☐ Correspondence address
- ☐ Application type: Non-Provisional Utility
- ☐ Priority claim: U.S. Provisional ⁶⁴/_{073,334}, filed May 23, 2026
- ☐ Related application: HEALTH-KEY ULTRA application number (to be inserted)
- ☐ Assignee: Embedded Operating Systems Research Foundation (EoS)

2. Specification — DRAFTED

- ☒ Title of Invention
- ☒ Cross-Reference to Related Applications
- ☒ Field of the Invention
- ☒ Background of the Invention

- ☒ Summary of the Invention
- ☒ Brief Description of the Drawings (7 figures described)
- ☒ Detailed Description of the Preferred Embodiments
- ☒ Claims (3 independent + 21 dependent = 24 total claims)
- ☒ Abstract (≤150 words)

File: HEALTH_BAND_Neuro_Utility_Patent_Application.md

3. Claims Summary

Claim	Type	Subject Matter
1	Independent	Zero-Hole Architecture / Pass-Through Power
2	Independent	Bidirectional Neuromuscular Interface (sEMG + TENS)
3	Independent	Integrated Breath Analysis Channel
4–6	Dependent on 1	USB-PD, composite USB enumeration, BLE
7–11	Dependent on 1	PPG, ECG, temperature, IMU, UV sensors
12–15	Dependent on 2	Electrode array, gesture ML, TENS parameters, isolation
16–20	Dependent on 3	PTFE membrane, Ti grille, fuel cell specs, VOC biomarkers, exhaust
21–23	Dependent on 1+2+3	Combined Zero-Hole apparatus, magnetic latch, hinge
24	Dependent on 1	Holographic micro-LED display (CIP seed claim)

Total claims: 24 (USPTO allows 20 claims with 3 independent in basic fee; claims 21–24 require excess claims fee)

4. Drawings — FORMAL DRAWINGS REQUIRED

- ☐ FIG. 1 — System block diagram (existing: `patent/figures/patent_fig1_block_diagram.png` — needs USPTO-compliant redraw)

- ☐ FIG. 2 — Exploded Core module view (existing: `patent/figures/patent_fig2_zero_hole_clasp.png` — needs USPTO-compliant redraw)
- ☐ FIG. 3 — Strap cross-section with electrode array (existing: `patent/figures/patent_fig3_semg_tens.png` — needs USPTO-compliant redraw)
- ☐ FIG. 4 — Breath analysis channel detail (new drawing required)
- ☐ FIG. 5 — Pass-through power schematic (new drawing required)
- ☐ FIG. 6 — Assembled wrist-worn perspective view (new drawing required)
- ☐ FIG. 7 — Neuromuscular interface circuit diagram (new drawing required)

USPTO Drawing Requirements:

- Black and white line drawings on 8.5" × 11" paper
- 1" margins on all sides
- Figure numbers and reference numerals in 12pt Arial or similar
- No shading (cross-hatching acceptable for cross-sections)
- Professional patent illustrator recommended

5. Oath or Declaration — USPTO Form PTO/AIA/08

- ☐ Inventor signature
- ☐ Date of signature
- ☐ Statement that inventor is the original inventor

6. Filing Fee (USPTO Fee Schedule, as of 2026)

Fee Item	Small Entity	Micro Entity
Basic filing fee (electronic)	\$320	\$160
Search fee	\$660	\$330
Examination fee	\$760	\$380
Excess claims (claims 21–24, 4 claims × \$50)	\$200	\$100
Total (estimated)	\$1,940	\$970

Note: Verify current fee schedule at [USPTO.gov](https://www.uspto.gov/fees) before filing. Micro entity status requires income certification.

7. Information Disclosure Statement (IDS) — USPTO Form PTO/SB/08

- ☐ List all known prior art references
- ☐ Include prior art cited in provisional application
- ☐ Include any references found during prior art search

Key Prior Art to Disclose:

- Apple Watch Series (PPG, ECG wearable)
- Fitbit Sense (EDA/GSR wearable)
- BACtrack Skyn (wrist-worn BAC sensor — different architecture)
- NeuroSky MindBand (EEG wearable)
- Myo Armband (sEMG gesture control — discontinued)
- TENS 7000 (standalone TENS unit)
- Anker PowerLine USB-C pass-through cables (pass-through charging — different form factor)

8. Assignment Document (if filing in assignee name)

- ☐ Assignment from inventor to Embedded Operating Systems Research Foundation (EoS)

- ☐ Notarized signature of inventor
- ☐ Recorded with USPTO Assignment Division

CIP Filing Checklist (Separate Filing — Due by May 23, 2027)

CIP New Matter to Add

- ☐ Holographic micro-LED display architecture (Claim 24 seed)
- ☐ Flexible polyimide micro-LED substrate specification
- ☐ Holographic diffuser film optical properties
- ☐ Bezel-less integration manufacturing process

CIP Strategy Document: `patent/cip/HEALTH_BAND_Neuro_CIP_Strategy.md`

Dual-Patent Portfolio Summary

Patent	Application No.	Filed	Status	Core Claims
HEALTH-KEY ULTRA	[To be assigned]	2026	CIP filed	USB-C dongle, Venturi breath channel, USB-C shield ECG
HEALTH-BAND Neuro	⁶⁴ / _{073,334} (provisional)	May 23, 2026	Non-provisional pending	Zero-Hole clasp, sEMG+TENS, wrist breath channel

This dual-patent strategy creates a comprehensive IP perimeter: competitors cannot copy the standalone dongle (Patent 1) nor the method of docking a diagnostic dongle into a gesture-control wristband (Patent 2).

Attorney Engagement Checklist

- ☐ Engage registered USPTO patent attorney or agent
- ☐ Provide attorney with this specification and all figures
- ☐ Attorney to conduct formal prior art search (FTO analysis)
- ☐ Attorney to finalize claim language and file formal application
- ☐ Attorney to prepare formal USPTO-compliant drawings
- ☐ Attorney to file IDS with all prior art references

Recommended Patent Firms (specializing in wearable/medical devices):

- Fish & Richardson P.C. (Boston/San Diego)
- Knobbe Martens (Irvine, CA — medical device specialty)
- Wilson Sonsini Goodrich & Rosati (Palo Alto)

Prepared by: Embedded Operating Systems Research Foundation (EoS)

Date: May 27, 2026